PEAD Strategy Showcase

Performance of all 8 tradeable strategy constructions, 1996–2013

Companion to *PEAD_Report.pdf* (evidence the effect exists) and *PEAD_Strategy_Development.pdf* (why each design decision was made)

This report answers one question for each of the 8 strategies built in this project: **if you had traded this exact rule from 1996 to 2013, what would you have earned, and at what risk?** Every number here is net of estimated transaction costs and realistic position-size limits (a stock can't be traded past 1-5% of its own daily volume), and every strategy compounds its own trailing net asset value (NAV) — sizing positions off a growing or shrinking capital base as the strategy wins and loses, not a fixed starting amount. All returns are **market-adjusted** (the return on the S&P/CRSP market index is subtracted out) except Strategy 7, which is explicitly built to let some market exposure through and so is reported in raw terms — see that strategy's own section for why.

A note on how these Sharpe ratios are calculated. Every strategy's daily return is computed as that day's dollar profit or loss divided by the *prior day's NAV*, not a fixed starting capital figure. An earlier version of this backtest divided by the fixed starting capital instead; since NAV compounds upward over a profitable 18-year run, that understated later years' percentage moves and made volatility look larger (and Sharpe smaller) than it really was. The numbers in this report use the corrected, NAV-based calculation throughout.

Overview: all 13 strategies side by side

Strategy	Ann. Return	Ann. Vol	Sharpe	Max Drawdown	Final NAV ($10M start)
1. Extreme decile L/S	4.91%	5.32%	0.93	-18.5%	$25.4M
2. Rank-weighted	3.88%	3.86%	1.01	-14.0%	$21.0M
3. Balanced (size-neutral)	4.74%	5.26%	0.91	-17.5%	$24.6M
4. Trimmed/tilted, priority cap	5.26%	4.41%	1.18	-13.1%	$27.1M
5. + Sector-neutral	4.06%	3.02%	1.33	-10.8%	$21.7M
6. + Leverage cap 2.5x	6.49%	4.94%	1.30	-16.6%	$34.0M
6+beta. + 0.5x market beta overlay	10.08%	10.51%	0.97	-23.2%	$64.9M
7. Unconstrained net exposure	7.30%	6.56%	1.11	-15.5%	$39.4M
6+beta-vt. Vol-targeted overlay	11.00%	10.66%	1.03	-18.5%	$76.3M
6-vt-alpha. Vol-targeted alpha only	10.74%	10.78%	1.00	-24.4%	$72.9M
6-fully-vt. Fully vol-managed	11.58%	10.86%	1.06	-17.9%	$84.5M
8. + Dispersion risk filter	5.96%	4.54%	1.30	-10.8%	$30.9M
8+beta-vt. Strategy 8 + overlay	10.58%	10.84%	0.98	-20.0%	$70.9M

Table 1. All figures 1996-2013, trailing-NAV compounding, net of transaction costs and liquidity caps, from a $10,000,000 starting capital base. Strategy 5 has the best Sharpe and shallowest drawdown among the original 8; Strategy 6-fully-vt now has the highest absolute return (see the "Extension" section below); Strategy 8 (second-to-last row, a separate follow-up research pass -- see "Extension 2" below) has the shallowest drawdown of any strategy in this table at -10.8%, with a Sharpe matching the best of the original 8 -- see the Recommendation section at the end of this report.

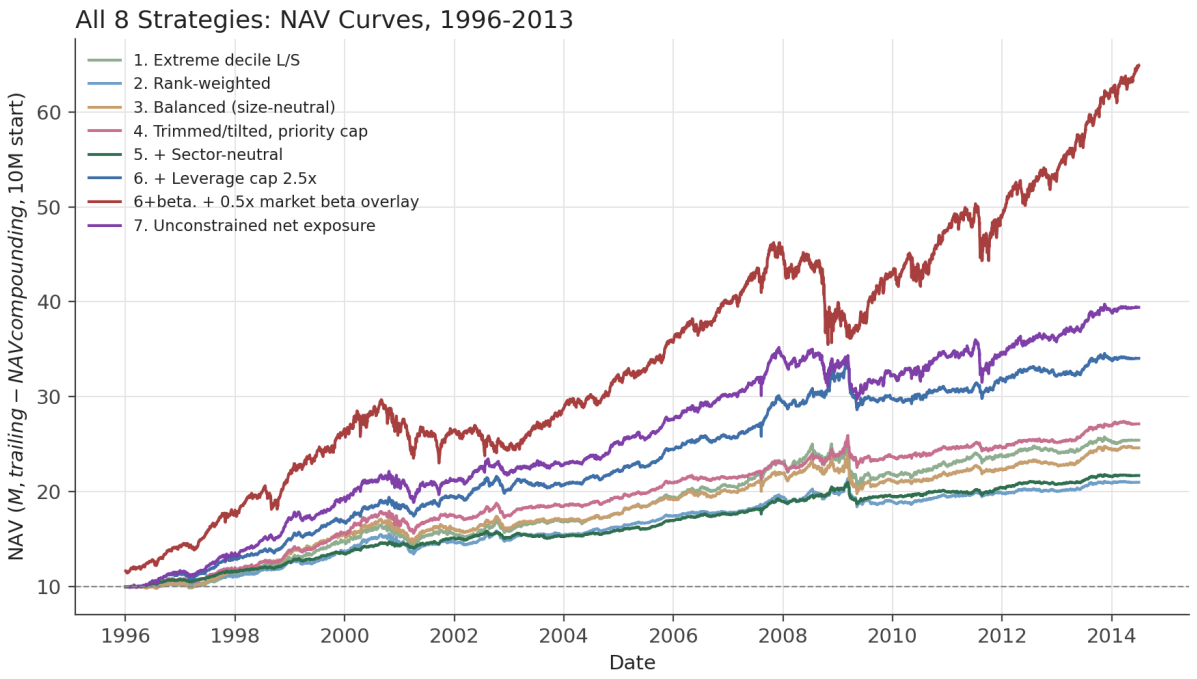


Figure 1. NAV curves for all 8 strategies. The beta-overlay strategy (dark red) pulls well ahead because it deliberately carries market exposure on top of the alpha book; the other 7 stay dollar-neutral (or close to it) by design and grow far more slowly but far more smoothly.

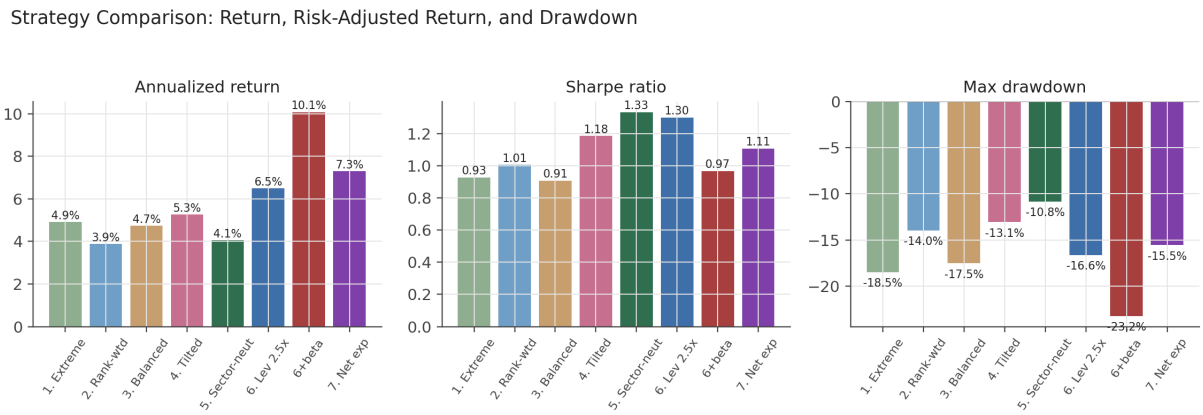


Figure 2. Return, Sharpe ratio, and max drawdown compared directly. Notice that higher return (Strategy 6+beta, Strategy 7) does not mean higher Sharpe -- Strategy 5 has the lowest return of the six core (non-overlay) strategies but the best risk-adjusted return and the shallowest drawdown, because it carries the least uncompensated risk.

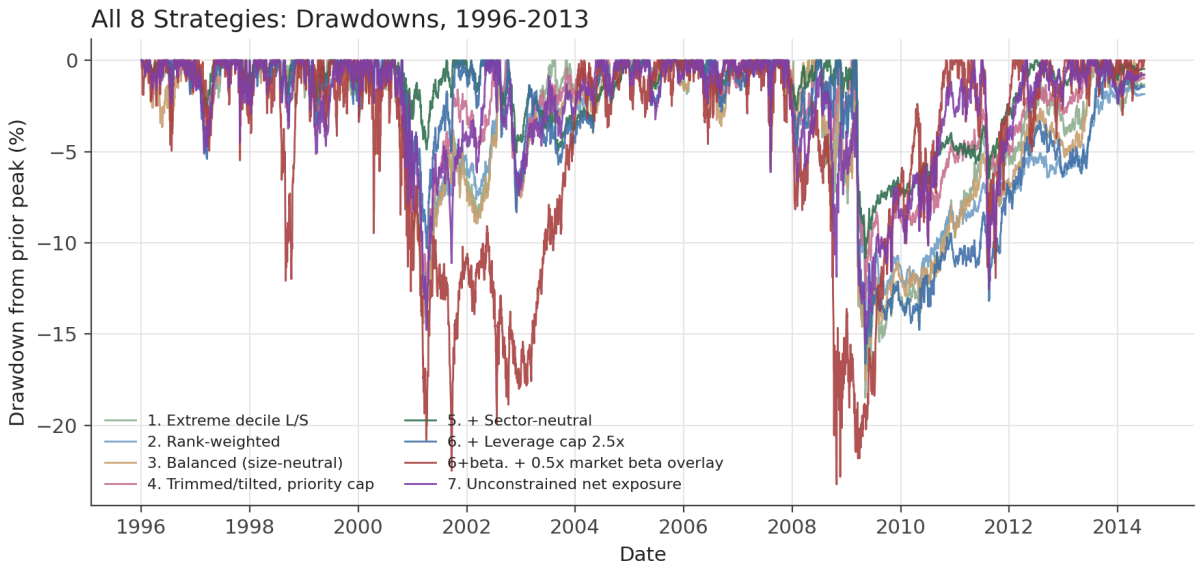


Figure 3. Drawdown from the prior peak, all 8 strategies. Every strategy's worst drawdown clusters around the 2008-2009 financial crisis. Strategy 6+beta's drawdown (-23.2%) is the deepest because it is the only strategy carrying meaningful net market exposure by construction.

Strategy profiles

Each profile below states, in plain terms, exactly how positions were chosen and sized, followed by the strategy's own metrics. See *PEAD_Strategy_Development.pdf* for the reasoning that led from one strategy to the next.

1. Extreme decile long/short

Long every stock in the most positive-surprise decile (D10), short every stock in the most negative-surprise decile (D1), equal +/-1 weight. The holding period is not a fixed number of days -- it comes from where each decile's own return curve was measured to flatten out (roughly 20-80 trading days depending on the decile; see *PEAD_Strategy_Development.pdf*'s Stage 1 for how this was decided from the data). This is the simplest possible way to trade the effect, and the baseline every later strategy is compared against.

Metric	Value
Annualized return	4.91%
Annualized volatility	5.32%
Sharpe ratio	0.93
Max drawdown	-18.5%
Avg. gross leverage	2.21x
Avg. concurrent positions	458
% positions liquidity-capped	35.0%
Total transaction costs	$3,476,377

The starting point. Profitable and sensible, but trades only the most extreme 20% of events and makes no attempt to control for sector or size tilts in the two legs.

2. Rank-weighted

Every event, across all ten deciles (not just the extremes), gets a position weight that scales continuously with how extreme its surprise rank was that quarter -- close to zero near the median, full weight at the tails. This uses the entire cross-section instead of discarding 80% of events at a hard cutoff.

Metric	Value
Annualized return	3.88%
Annualized volatility	3.86%
Sharpe ratio	1.01
Max drawdown	-14.0%
Avg. gross leverage	2.01x
Avg. concurrent positions	1,934
% positions liquidity-capped	99.3%
Total transaction costs	$3,577,068

Better Sharpe than the extreme-decile version (more diversification, lower volatility) but lower absolute return, and it saturates the liquidity cap on 99% of its positions and the leverage cap in 70 of its quarters -- trading this many names runs into real capacity limits. This tradeoff is exactly what motivated Strategy 4.

3. Balanced (size-neutral)

Same D10/D1 extreme-decile selection as Strategy 1, but each side's weight is redistributed across firm-size quintiles every quarter so no single size bucket dominates either leg. This guards against the long and short legs accidentally differing in average firm size, which would secretly turn part of the return into an unrelated size bet rather than a PEAD bet.

Metric	Value
Annualized return	4.74%
Annualized volatility	5.26%
Sharpe ratio	0.91
Max drawdown	-17.5%
Avg. gross leverage	2.02x
Avg. concurrent positions	458
% positions liquidity-capped	18.3%
Total transaction costs	$3,158,625

A direct, targeted fix for one specific risk (size imbalance), not a redesign of the whole strategy. Performs close to Strategy 1, which is itself informative: size imbalance was not, on its own, doing much of the work in Strategy 1's return.

4. Trimmed / tilted, priority-based cap

Built after diagnosing exactly why rank-weighted's Sharpe was better but its return worse: it was trading about 4x more positions than the extreme-decile strategies, hitting the liquidity cap on 99% of positions and the leverage cap in 90% of quarters, and the old leverage cap trimmed *every* position proportionally when it bound -- diluting the highest- and lowest-conviction trades equally. Strategy 4 trims out the low-conviction middle of the SUE-rank distribution entirely, tilts remaining weight toward the extremes, keeps size-neutrality, and replaces proportional cap-trimming with **priority-based** trimming: when the leverage cap binds, the lowest-conviction positions are dropped first, not everyone shaved by the same percentage.

Metric	Value
Annualized return	5.26%
Annualized volatility	4.41%
Sharpe ratio	1.18
Max drawdown	-13.1%
Avg. gross leverage	2.49x
Avg. concurrent positions	1,406
% positions liquidity-capped	8.5%
Total transaction costs	$4,456,823

The first strategy to meaningfully beat the v1 baseline on both return (5.26% vs. 4.91%) and Sharpe (1.18 vs. 0.93) at the same time, rather than trading one for the other.

5. + Sector-neutral

A diagnostic run on Strategy 4 found its book concentrating up to **29% of gross weight in a single Fama-French sector** in some quarters (versus 8.3% if evenly spread across all 12 sectors) -- an unpriced, PEAD-unrelated risk the strategy was carrying by accident. Strategy 5 adds sector-neutrality alongside the existing size-neutrality, redistributing weight within each sector the same way size was already balanced within each quintile.

Metric	Value
Annualized return	4.06%
Annualized volatility	3.02%
Sharpe ratio	1.33
Max drawdown	-10.8%
Avg. gross leverage	1.59x
Avg. concurrent positions	1,406
% positions liquidity-capped	3.1%
Total transaction costs	$2,848,909

The best risk-adjusted strategy built in this project: Sharpe 1.33, max drawdown only -10.8% (the shallowest of all 8). Removing an unpriced risk, rather than chasing more return, is what produced the best risk-adjusted outcome -- see Figure 4 in the development report for the concentration data behind this decision.

6. + Leverage cap raised 1.5x -> 2.5x

Tests whether Strategy 5's 1.5x gross leverage cap was actually costing it anything, now that sector concentration (the risk the cap was implicitly protecting against) is gone. A sweep of the cap from 1.5x up to 10x found returns and Sharpe both flatten out entirely by about 2.0-2.5x -- the point where the cap stops binding at all for this specific, well-diversified signal. Strategy 6 raises the cap to 2.5x.

Metric	Value
Annualized return	6.49%
Annualized volatility	4.94%
Sharpe ratio	1.30
Max drawdown	-16.6%
Avg. gross leverage	4.00x
Avg. concurrent positions	1,406
% positions liquidity-capped	7.2%
Total transaction costs	$6,845,799

Improved both return (5.08%->6.49%) *and* Sharpe versus Strategy 5, because there was no longer any concentration risk left for the tighter cap to be protecting against. This is the recommended core (pure-alpha) strategy for live consideration.

6+beta. Strategy 6 + 0.5x market beta overlay

Strategy 6's alpha book is left completely untouched. A separate synthetic market-index position, sized to 0.5x the strategy's own trailing NAV and rebalanced on the same quarterly cadence (2bps overlay transaction cost), is held *alongside* it -- the standard institutional way (a "130/30"-style long-extension structure) to add back market exposure without distorting the alpha signal itself.

Metric	Value
Annualized return	10.08%
Annualized volatility	10.51%
Sharpe ratio	0.97
Max drawdown	-23.2%

Highest absolute return of any strategy built (10.08%), landing inside a 10-15% equity-market-like target band, with a much shallower drawdown and better Sharpe than simply holding the market outright over the same period. Recommended if a return closer to the market's, with correspondingly more market-correlated risk, is preferred over a pure, uncorrelated alpha stream.

7. Unconstrained net exposure

Tests the *other* way to let market exposure through: instead of adding a separate overlay, stop forcing the long leg's and short leg's dollar totals to match every quarter (the size/sector-neutral construction does this by construction). Removing that constraint required switching from market-adjusted to **raw** returns -- market-adjusting would have silently cancelled out the very net exposure this test is trying to let through.

Metric	Value
Annualized return	7.30%
Annualized volatility	6.56%
Sharpe ratio	1.11
Max drawdown	-15.5%
Avg. concurrent positions	1,406
% positions liquidity-capped	7.9%
Total transaction costs	$7,413,491
Correlation to market	0.52

A cautionary result, not a recommended design: despite averaging only -5.5% net exposure as a fraction of gross (essentially flat by that measure), Strategy 7's correlation to the market came out at **0.52** -- much higher than intended -- because dollar-neutral is not the same thing as beta-neutral. Positive- and negative-surprise firms structurally differ in average beta, so equal dollar amounts long and short don't cancel market risk the way market-adjusting the underlying returns does. Kept in the repo as a fully documented, tested result precisely because it demonstrates that lesson.

Extension: volatility-targeted risk management

A follow-up research pass asked whether Strategy 6 + 0.5x beta overlay's return could be pushed higher without simply trading away Sharpe and drawdown -- i.e. without just raising the overlay's size or the alpha book's leverage, both of which were already shown (the leverage-cap sweep behind Strategy 6, and a base_unit_fraction sizing sweep re-run on this specific book) to trade return for materially worse Sharpe and drawdown past their respective saturation points, not to be a free improvement. The alternative tested here is **volatility targeting** (Moreira & Muir 2017-style volatility-managed exposure): instead of holding a *fixed* notional every quarter, scale it by (a target volatility level) / (trailing realized volatility), so the position is sized *down* in turbulent quarters and *up* in calm ones, clipped to [0.5x, 2.0x] of its baseline sizing. The target level is calibrated to the sample's own average trailing volatility, so average exposure stays comparable to the static baseline -- this keeps the comparison a genuine risk-timing question, not a comparison against a different average market exposure.

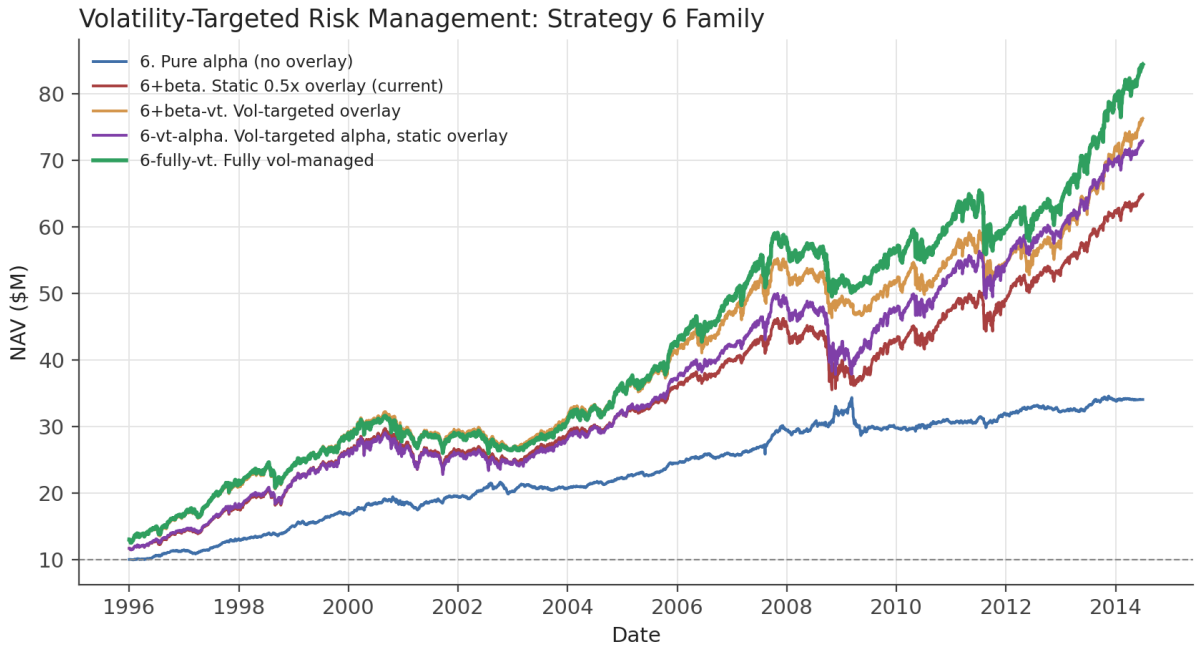


Figure 5. NAV curves for the Strategy 6 family: pure alpha, the current static 0.5x overlay, and the three vol-targeted variants. The fully vol-managed variant (green, thickest line) tracks above the static overlay for most of the sample and, critically, pulls up out of the 2008-2009 drawdown faster.

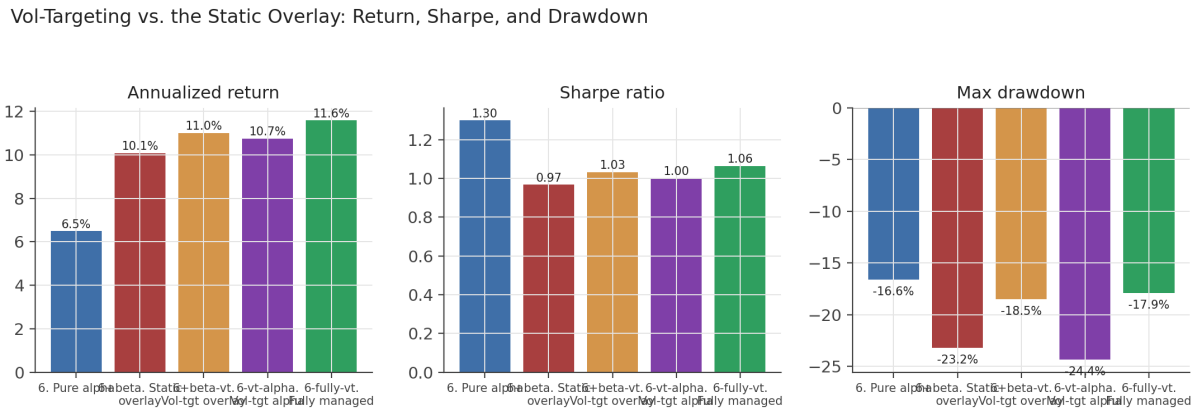


Figure 6. Unlike raising the overlay's static size or the alpha book's leverage (both shown elsewhere in this project to trade return for worse Sharpe/drawdown), vol-targeting improves all three metrics at once for the fully-managed variant.

6+beta-vt. Vol-targeted 0.5x beta overlay

Same construction as Strategy 6+beta -- the alpha book is untouched, a separate synthetic index position is held alongside it -- except the overlay's notional is scaled each quarter by trailing 63-day realized market volatility relative to its own full-sample average, instead of being pinned to a fixed 0.5x NAV every quarter.

Metric	Value
Annualized return	11.00%
Annualized volatility	10.66%
Sharpe ratio	1.03
Max drawdown	-18.5%

Beats the static overlay on every metric: 11.00% return (vs. 10.08%), Sharpe 1.03 (vs. 0.97), max drawdown -18.5% (vs. -23.2%). The overlay delevers heading into 2008 instead of holding a fixed market-sized position through it.

6-vt-alpha. Vol-targeted alpha book, static overlay

Isolates the other half of the idea: leave the beta overlay static, and instead scale the alpha book's own exposure each quarter by the same trailing-volatility logic, applied to the alpha book's own realized vol rather than the market's. (Approximation: this rescales the already-computed daily return series rather than re-running the position-level engine at a different size, so it does not reproduce exactly how the liquidity cap would respond, and charges no extra cost for the implied rebalancing beyond what's already embedded in the return series.)

Metric	Value
Annualized return	10.74%
Annualized volatility	10.78%
Sharpe ratio	1.00
Max drawdown	-24.4%

Improves return (10.74%) and Sharpe (1.00) over the static baseline, but on its own makes the drawdown slightly *worse* (-24.4% vs. -23.2%) -- vol-targeting the alpha book alone apparently increases exposure at some points that coincide with the overlay's own worst stretch. This nuance is exactly why the combined variant below, not this one alone, is the one worth adopting.

6-fully-vt. Fully vol-managed (vol-targeted alpha + overlay)

Both halves combined: the alpha book and the beta overlay are each scaled quarterly by their own trailing-volatility signal (same [0.5x, 2.0x] clipping, same calibration-to-average discipline as the two variants above).

Metric	Value
Annualized return	11.58%
Annualized volatility	10.86%
Sharpe ratio	1.06
Max drawdown	-17.9%

The best result of this research pass, and the new top-return recommendation: **11.58% return, Sharpe 1.06, max drawdown -17.9%** -- better than the static-overlay baseline on every dimension (10.08% / 0.97 / -23.2%), and resolving the drawdown regression seen in the alpha-only variant above. See *PEAD_Strategy_Development.pdf* Stage 7 for the full reasoning and the sizing-sweep diagnostic that ruled out simple leverage as the answer.

Extension 2: a dispersion-based risk filter (Strategy 8)

A second follow-up research pass asked whether any signal outside SUE itself -- genuinely new information, not another way to size or overlay the existing signal -- could improve the strategy. Three candidates were tested from data already on hand (no new pull): a second earnings-surprise measure (sue_seasonal, a seasonal-random-walk expectation already computed in this project's own data), analyst estimate revisions and dispersion (from IBES's own consensus summary file, untouched until now), and a revenue surprise built from Compustat quarterly revenue (also untouched until now). Each was tested the same rigorous way as the EAR and SUE-reversal dead ends earlier in this project: a standalone Fama-MacBeth predictive check, then an actual blended-weight trading test through the full position-level engine -- not just statistical significance in isolation.

Two of the three failed outright despite real literature precedent and, in sue_seasonal's case, standalone statistical significance (t=3.25): blending either one into the strategy's weight construction made return, Sharpe, *and* drawdown monotonically worse at every blend weight tested. Revenue surprise -- the best-precedented candidate on paper (Jegadeesh & Livnat 2006) -- was only marginally significant standalone (t=1.47) and likewise did not survive the blend test. All three are documented negative results, in the same spirit as the EAR and SUE-reversal dead ends.

Analyst estimate dispersion produced a real but counterintuitive result. Standalone, it has *no* directional predictive power at all -- a quintile sort of forward returns by dispersion alone is flat (t=-0.43). But blending it into the trim/tilt weight construction at a modest weight changes which low-conviction positions get trimmed: Sharpe stays in a ~1.28-1.32 plateau across blend weights 0.05-0.25 while max drawdown steadily improves from -17.3% to -9.4%. A composition check (average size quintile, liquidity quintile, dollar volume) found this is *not* a disguised size or liquidity filter -- those are essentially unchanged. The most plausible mechanism: dispersion measures how much analysts disagree about the estimate a surprise is being measured against, so a high-dispersion event's "surprise" is measured against a noisier consensus and is more likely to be noise than real information. **This is a risk/variance filter, not a new alpha signal** -- the same kind of finding as Strategy 5's sector-neutrality (better risk-adjusted profile, not more expected return), and it is described that way deliberately rather than oversold.

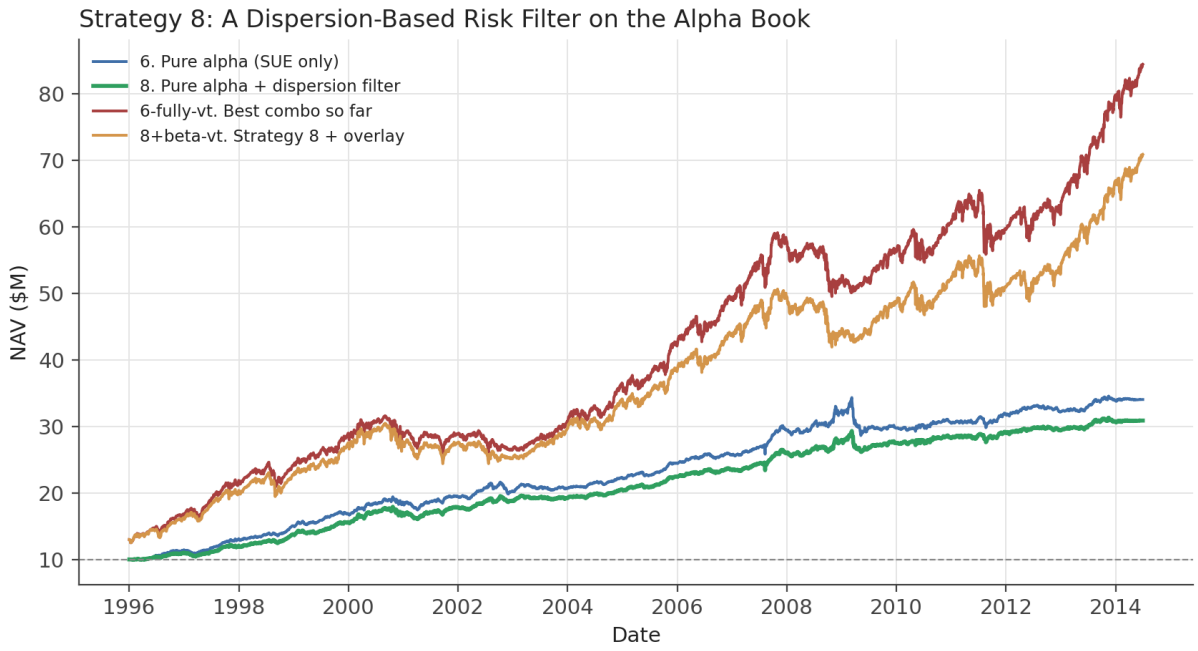


Figure 7. Strategy 8 (green) tracks below Strategy 6 in absolute NAV but with visibly smaller pullbacks throughout, including 2008-2009. Strategy 8 + the vol-targeted overlay (orange) sits below Strategy 6-fully-vt (red) for most of the sample.



Figure 8. Strategy 8 gives up 0.36pp of return relative to Strategy 6 for a Sharpe that matches or beats it and a drawdown less than two-thirds as deep. Combining it with the vol-targeted overlay (rightmost bars) does not improve on Strategy 6-fully-vt.

8. Pure alpha + dispersion risk filter

Identical to Strategy 6's construction (trim=0.15, tilt_power=1.0, size+sector-neutral, priority-based leverage cap at 2.5x) except the conviction rank used for trim/tilt/sizing blends sue_rank_pct (weight 0.80) with a point-in-time-safe analyst-dispersion rank (weight 0.20, from IBES's own consensus summary, using the latest snapshot strictly before the announcement).

Metric	Value
Annualized return	5.96%
Annualized volatility	4.54%
Sharpe ratio	1.30
Max drawdown	-10.8%

5.96% return, Sharpe 1.30, max drawdown -10.8% -- the shallowest drawdown of any strategy in this project, at a Sharpe matching Strategy 6's. A genuine alternative pure-alpha recommendation to Strategy 6, not a replacement for it: choose based on whether the extra 0.5pp of Strategy 6's return is worth carrying a materially deeper drawdown.

8+beta-vt. Strategy 8 + vol-targeted beta overlay

The natural next question: do Strategy 8's drawdown improvement and the vol-targeted overlay's drawdown improvement (Extension 1, above) compound when combined? Same overlay construction as Strategy 6+beta-vt, laid on top of Strategy 8's alpha book instead of Strategy 6's.

Metric	Value
Annualized return	10.58%
Annualized volatility	10.84%
Sharpe ratio	0.98
Max drawdown	-20.0%

They do not compound -- they don't even hold up separately. 10.58% return, Sharpe 0.98, max drawdown -20.0%, worse than Strategy 6-fully-vt (11.58% / 1.06 / -17.9%) on every metric, despite Strategy 8 alone having a much shallower drawdown than Strategy 6 alone. Reported as tested: this combination is not recommended. Strategy 8's value is as a pure-alpha choice on its own, not as an ingredient to combine with the overlay.

Recommendation

For live consideration: Strategy 6 or Strategy 8 for pure alpha (choose based on return-vs-drawdown preference), or Strategy 6-fully-vt if a higher, more market-like return is preferred. Every strategy in this project is deliberately built market-adjusted and (mostly) dollar-neutral, which means it structurally forgoes the equity risk premium that makes up most of a simple buy-and-hold return -- comparing its nominal return to the S&P 500 directly is not apples to apples. As an external sanity check, AQR's live Equity Market Neutral Fund (QMNNX) returns 6.19% at 7.03% volatility (Sharpe 0.69) and also trails the S&P 500 over the same window -- that is what a professionally run market-neutral strategy is supposed to look like. Strategy 6 already beats that real fund's Sharpe (1.30 vs. 0.69), and Strategy 8 beats it further on drawdown.

• Choose **Strategy 6** for the highest pure-alpha return (6.49%, Sharpe 1.30, max drawdown -16.6%), or **Strategy 8** if a shallower drawdown matters more than that last 0.5pp of return (5.96%, Sharpe 1.30, max drawdown -10.8%, the shallowest in this project) -- both close to zero market beta by construction.
• Choose **Strategy 6-fully-vt** if a return closer to 10-15% is preferred: it remains the best return-oriented choice after testing whether Strategy 8 could improve on it in combination (it doesn't -- see Extension 2) -- beating the original static overlay on return (11.58% vs. 10.08%), Sharpe (1.06 vs. 0.97), *and* max drawdown (-17.9% vs. -23.2%) simultaneously.
• The plain static **Strategy 6+beta**, **Strategy 7**, and **Strategy 8+beta-vt** are superseded/not recommended: the vol-targeted overlay dominates the static one on every metric, Strategy 7 adds market exposure in a way that is measurably less controllable than the explicit overlay, and combining Strategy 8 with the overlay underperforms Strategy 6-fully-vt on every metric despite Strategy 8's own strong standalone drawdown profile.

Caveats specific to this backtest

• **Transaction costs and liquidity caps are disclosed assumptions, not calibrated figures.** The cost schedule and position-size caps are reasonable, conservative-leaning choices, not fitted to any real broker, venue, or historical execution data.
• **No borrow cost or short-availability modeling.** Every strategy that shorts D1 assumes the short is executable at the modeled cost with no separate borrow fee and no risk the shares simply aren't available to borrow -- a real constraint, worse for exactly the small, illiquid names that dominate the negative-surprise decile.
• **One historical realization, not walk-forward validated.** Each strategy reports one full-sample backtest over 1996-2013; none of the design choices (holding-period buckets, cost tiers, leverage cap, trim/tilt parameters) were tuned on a held-out period and validated out-of-sample separately.
• **Data ends in 2013.** PEAD is a well-known, heavily studied anomaly; a live strategy today would need current data revalidation, and this project's own era-subsample analysis (see *PEAD_Report.pdf* Section 5c) already found weaker point estimates in the most recent years of this sample.

Generated from data/results_summary_v2_FINAL.csv and data/backtest_v2_*_summary.json. Scripts: scripts/14_daily_decay.py through scripts/26_strategy7_unconstrained_netexposure.py, scripts/33_strategy6_beta_overlay_voltarget.py, scripts/34_strategy6_voltarget_combined.py, scripts/35_strategy6_sizing_sweep.py, scripts/36_revision_dispersion_diagnostic.py, scripts/37_revenue_surprise_diagnostic.py, scripts/38_sue_seasonal_diagnostic.py, scripts/39_strategy8_dispersion_filter.py, scripts/40_strategy8_beta_overlay_voltarget.py, scripts/29_v2_strategy_charts.py, scripts/30_build_strategy_showcase_report.py.